EXP-15 : To rank Caesar-cipher decryptions by English letter frequency.

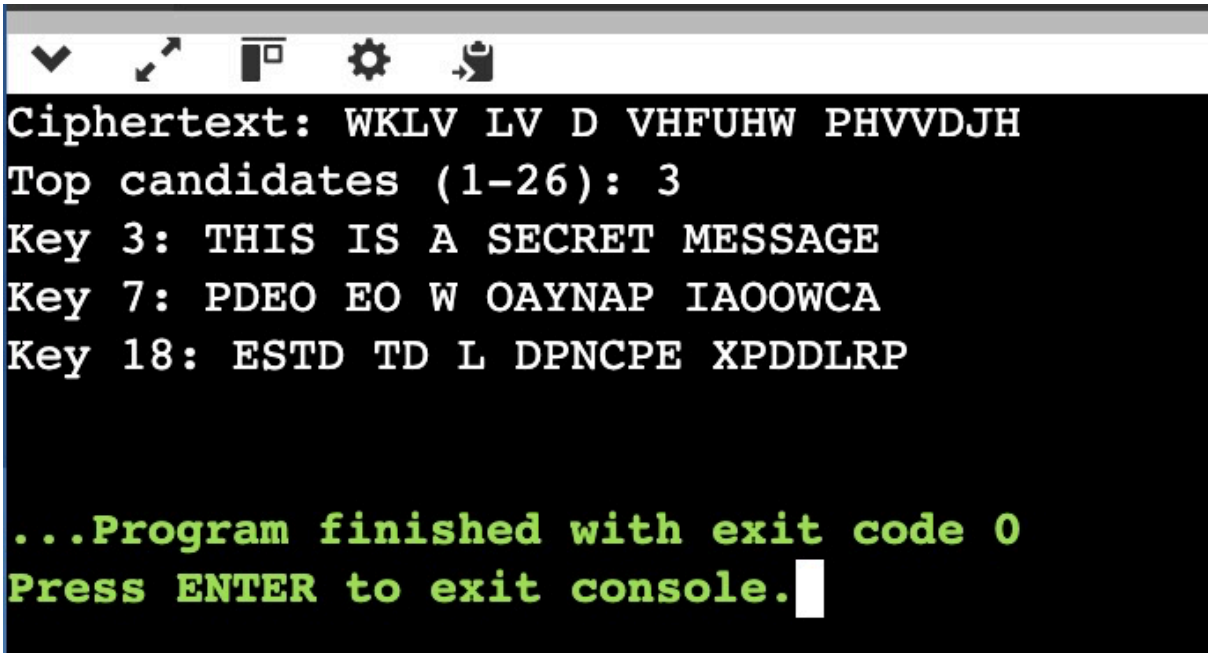
PROGRAM :

Run Debug Stop Share Save { } Beautify

main.c

```c
#include <stdio.h>
#include <ctype.h>
int main(void) {
char c[501]; int top, order[26]; double score[26];
double f[] = {8.167,1.492,2.782,4.253,12.702,2.228,
2.015,6.094,6.966,.153,.772,4.025,2.406,6.749,
7.507,1.929,.095,5.987,6.327,9.056,2.758,.978,
2.360,.150,1.974,.074};
printf("Ciphertext: ");
if (!fgets(c,sizeof c,stdin)) return 1;
printf("Top candidates (1-26): ");
if (scanf("%d", &top)!=1 || top<1 || top>26) return 1;
for (int k = 0; k < 26; k++) {
int count[26] = {0}, n = 0;
for (int i = 0; c[i]; i++) {
int x = toupper((unsigned char)c[i]);
if (x >= 'A' && x <= 'Z') {
count[(x-'A'-k+26)%26]++; n++;
}
}
if (!n) { puts("No letters."); return 1; }
score[k] = 0; order[k] = k;
for (int j = 0; j < 26; j++) {
double expected = n*f[j]/100;
double d = count[j]-expected;
score[k] += d*d/expected;
}
}
for (int i = 0; i < 25; i++)
for (int j = i+1; j < 26; j++)
if (score[order[j]] < score[order[i]]) {
int t=order[i]; order[i]=order[j]; order[j]=t;
}
for (int j = 0; j < top; j++) {
int k = order[j]; printf("Key %d: ",k);
for (int i = 0; c[i]; i++) {
int x = toupper((unsigned char)c[i]);
putchar(x>='A' && x<='Z' ? (x-'A'-k+26)%26+'A' : x);
}
}
return 0;
}
```

OUTPUT :


Ciphertext: WKLV LV D VHFUHW PHVVDJH
Top candidates (1-26): 3
Key 3: THIS IS A SECRET MESSAGE
Key 7: PDEO EO W OAYNAP IAOOWCA
Key 18: ESTD TD L DPNCPE XPDDLRP
...Program finished with exit code 0
Press ENTER to exit console.